

Answers

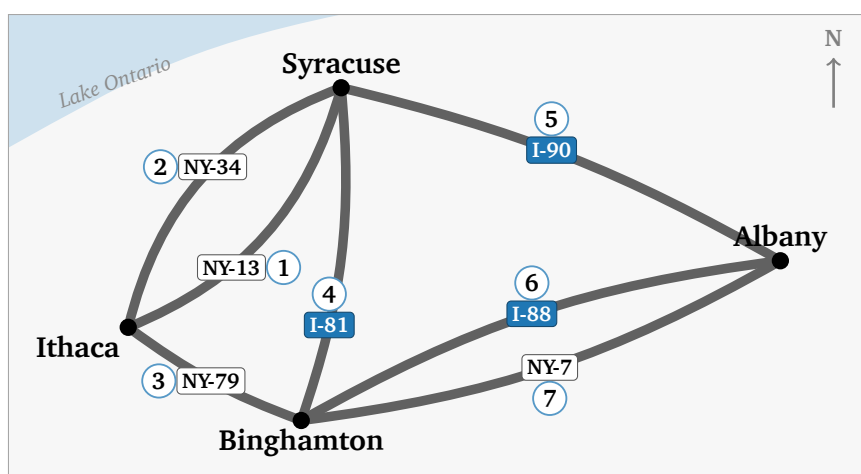
The sheet with its answers, **in blue**. Question 1(a) has many right answers and one of them is drawn; every other question has one.

Part 1 — Seven highways

Seven highways join four Upstate New York cities: Ithaca, Syracuse, Binghamton, Albany.

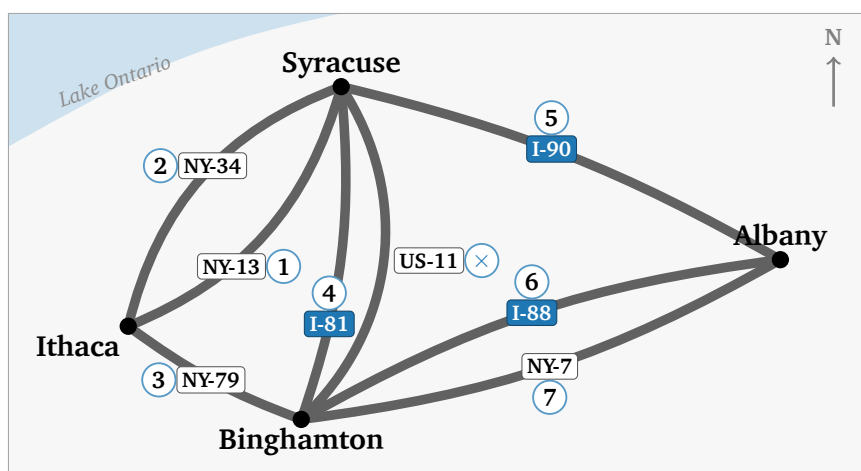
Question 1(a): Drive every highway exactly once, without lifting your pencil. Start and finish anywhere; cities may repeat, highways may not. Write in each circle when you drive that road — 1 first, then 2, on to 7.

⇒ One drive out of many. Whichever roads it takes in between, it has to start at **Ithaca** and finish at **Albany**: those are the two cities with an odd number of roads, and an odd city is where a drive begins or ends.

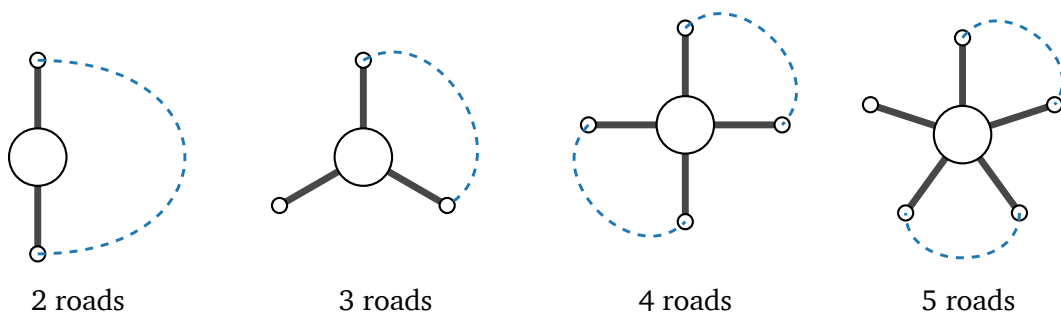


Question 1(b): The same four cities, and an eighth highway: US-11, which really does run alongside I-81. Number your route again. Can you drive every highway without lifting your pencil?

⇒ **No**. Seven of the eight can be driven — the numbers above still work — and **US-11 is left over**, or I-81 if you would rather leave that one. US-11 pushes Syracuse to five roads and Binghamton to five, so **four** cities now have a left-over road, and a drive has one start and one finish.



Question 2: One city, on its own. Arriving uses up one highway, leaving uses up another — so highways get used in **pairs**. Join two road-ends with a curved line to mean “in here, out there”. Pair up as far as you can.



	2 roads	3 roads	4 roads	5 roads
pairs you drew	1	1	2	2
roads left over	0	1	0	1

(a) Which ones can you drive straight *through*? $\Rightarrow$ **2 and 4**, the ones with nothing left over. (b) What do those numbers share? $\Rightarrow$ **They are even.** (c) A left-over road has no partner, so this city must be the tour's $\Rightarrow$ **start or finish.**

Question 3: Count the highways at each city.

	I	S	B	A	how many have a left-over?
seven roads	3	4	4	3	2
with US-11 too	3	5	5	3	4

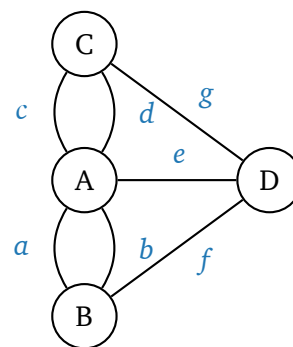
One start, one finish. So at most **2** cities may have a left-over road. Seven roads: **a drive exists** (Ithaca to Albany). Eight roads: **none exists**. No route with eight because **four cities have a left-over road, and a drive has only one start and one finish.**

Question 4: The same puzzle, 1736. Königsberg: four pieces of land, seven bridges. Two pairs join the same land; both of a pair must be crossed.

$\Rightarrow$ Bridges: **A 5, B 3, C 3, D 3** — fourteen ends for seven bridges, as always.

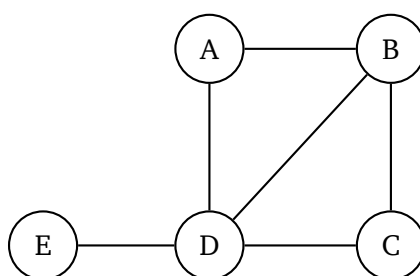
$\Rightarrow$ Is the walk possible? **No**: all four pieces of land are odd.

$\Rightarrow$ Destroying one bridge is enough. Which ones work? **Any of the seven.** Every bridge joins two odd lands, so knocking one out makes both of them even and leaves exactly two odd lands — and the map stays in one piece whichever you choose. (Königsberg's own answer was to build one, not to destroy one.)



Part 2 — Three kinds of route

A different network: five bus stops.



Question 5(a): Fill in the table. The last column takes one of these three names.

Repeats nothing: **path**. Never repeats a line: **trail**. Repeats both: **walk**.

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	walk
A B D C B	no	yes (B)	trail
A B C D E	no	no	path

Question 5(b): A route from A to E that repeats a stop but no line.

⇒ **A D B C D E**. The lines A–D, D–B, B–C, C–D, D–E are five different lines; D is the stop that comes round twice.

Question 5(c): Your drive in 1(a) used no road twice, but it did come back to a city it had already visited. It is a ...

⇒ **trail**. No road twice makes it a trail; coming back to a city it had already visited is what stops it being a path.

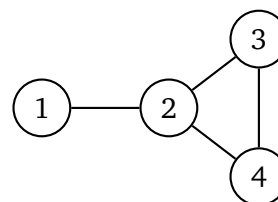
Question 5(d): It could not have been a path. A path of 7 roads would have to touch ... different cities, and the highway map has only ...

⇒ **8** cities — one more city than roads, because a path never comes back — and the map has only **4**.

Part 3 — Predictions for the notebook

Question 6(a): Write **1** in row i , column j if a line joins i and j , else **0**. **6(b):** Add up each row.

$A =$	1	2	3	4	row total
1	0	1	0	0	1
2	1	0	1	1	3
3	0	1	0	1	2
4	0	1	1	0	2



Each row total is the **degree** of that node.

Question 6(c): A **2-step route** goes along one line, then along one more. Count them on the picture.

⇒ From 1 to 3: **1** of them, $1 \rightarrow 2 \rightarrow 3$. From 1 to 2: **0**, because **the first step has to go to 2, and the second step has to leave it again**. From node 2 back to node 2: **3**, out and back along each of its three lines.

Question 6(d): The notebook will square your grid. Say what you expect to find in A^2 .

⇒ Row 1, column 3: **1**. Row 1, column 2: **0**. The diagonal is **the degrees** — 1, 3, 2, 2 — because a 2-step route that comes back is one line out and the same line home.

$$A^2 = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Part 4 — The lab

The seven roads, read off the map — the order does not matter, but both Ithaca–Syracuse roads must be there and both Binghamton–Albany roads must be there:

ROADS = [(0, 1), (0, 1), (0, 2), (1, 2), (1, 3), (2, 3), (2, 3)]

```

def to_adjacency(edges, n):
    A = np.zeros((n, n), dtype=int)
    for i, j in edges:
        A[i, j] += 1
        A[j, i] += 1
    return A

def degrees(A):
    return A.sum(axis=1)

def euler_status(A):
    if not is_connected(A):
        return "impossible"
    odd = int(np.sum(degrees(A) % 2 == 1))
    if odd == 0:
        return "circuit"
    if odd == 2:
        return "path"
    return "impossible"

```

The verdicts: seven roads **path**, with US-11 **impossible** — 1(a) and 1(b) again, out of a rule that never looked at the picture.

Finished early. Every degree even and still no tour means the map is **in more than one piece**: two triangles, say,

```
CHALLENGE = [(0, 1), (1, 2), (2, 0), (3, 4), (4, 5), (5, 3)]
```

Every place has two roads, and no drive gets from one triangle to the other. A rule that only counted parity says circuit here and is wrong; that is what `is_connected(A)` is for. One road joining the two triangles makes its two ends odd and the answer path; a second road *beside the first*, between the same two places, makes everything even again and the answer circuit.

The whole notebook with every blank filled in is `lab-solutions.py`, built from `lab.py` by `tools/build_m01_lab_notebooks.py`.